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Electromyography of Superficial and Deep Neck Muscles During Isometric, Voluntary, and Reflex Contractions

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Increasingly complex models of the neck neuromusculature need detailed muscle and kinematic data for proper validation. The goal of this study was to measure the electromyographic activity of superficial and deep neck muscles during tasks involving isometric, voluntary, and reflexively evoked contractions of the neck muscles. Three male subjects (28–41 years) *had electromyographic (EMG) fine wires inserted into the left sternocleidomastoid, levator scapulae, trapezius, splenius capitis, semispinalis capitis, semispinalis cervicis, and multifidus muscles. Surface electrodes were placed over the left sternohyoid muscle. Subjects then performed: (i) maximal voluntary contractions (MVCs) in the eight directions (*45 deg *intervals) from the neutral posture; (ii)* 50 N *isometric contractions with a slow sweep of the force direction through* 720 deg*; (iii) voluntary oscillatory head movements in flexion and extension; and (iv) initially relaxed reflex muscle activations to a forward acceleration while seated on a sled. Isometric contractions were performed against an overhead load cell and movement dynamics were measured using six-axis accelerometry on the head and torso. In all three subjects, the two anterior neck muscles had similar preferred activation directions and acted synergistically in both dynamic tasks. With the exception of splenius capitis, the posterior and posterolateral neck muscles also showed consistent activation directions and acted synergistically during the voluntary motions, but not during the sled perturbations. These findings suggest that the common numerical-modeling assumption that all anterior muscles act synergistically as flexors is reasonable, but that the related assumption that all posterior muscles act synergistically as extensors is not. Despite the small number of subjects, the data presented here can be used to inform and validate a neck model at three levels of increasing neuromuscular–kinematic complexity: muscles generating forces with no movement, muscles generating forces and causing movement, and muscles generating forces in response to induced movement. These increasingly complex data sets will allow researchers to incrementally tune their neck models' muscle geometry, physiology, and feedforward/feedback neuromechanics.* [DOI: 10.1115/1.2401185]

Keywords: muscle activity, kinematics, numerical model validation

Introduction

Neck muscles are increasingly included in physical and numerical models used to study traumatic neck injury. Early recognition of the importance of neck muscles on head dynamics resulted in anthropometric test dummy (ATD) response corridors being developed from tests of a pretensed human subject [1,2]. More recently, cadaveric spine preparations used to study traumatic neck injury have also added muscle forces to improve biofidelity [3–5]. Numerical models have shown that the inclusion of either passive or active neck musculature alters head kinematics during moderate speed collisions [6–9]. Active neck muscles have also been shown to alter the tensile load distribution in the cervical spine and shift the site of potentially catastrophic injuries from the upper to lower cervical spines [10,11]. Taken together, these studies indicate that neck muscles should be considered when studying traumatic neck injury.

Unlike bones, ligaments and intervertebral discs, musculotendinous tissues are active rather than passive structures. Muscles stiffness varies by an order of magnitude between the passive and maximally active states [12,13] and can be transiently increased even further when reflex activation is superimposed on tonically active muscles [14]. Muscle activation levels also vary rapidly during a collision: ranging from quiescent levels immediately before and after impact to near maximal voluntary levels within 100 ms of impact [15]. Moreover, each of the more than 20 pairs of neck muscles [16]—many of which can be subdivided into functionally distinct subvolumes—may have a different time-varying activation pattern. Thus the inclusion of neck muscles into a numerical model may improve its biofidelity, but it also increases its complexity.

Current physical models of the neck (e.g., ATDs) include neither individual muscles nor dynamic changes in muscle activation levels. Instead, the neck properties are adjusted to fit within relatively broad corridors [1]. Current cadaveric preparations include 2–5 simulated neck muscles [17,5], but do not yet simulate varying muscle activation during the dynamic response. Many numerical models have incorporated more anatomical detail and have included 20 or more muscle pairs [18,7,19,20,10,11,21]. Some numerical models include only the passive muscle response [22–24], whereas others use a constant maximal or submaximal activation strategy [25,26,8,20,10] or reflexively activated muscles [6,7,9,21]. Despite this complexity, most numerical models with active muscles lump anterior and posterior muscles into

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Table 1 Subject information

Subject	1	2	3
Age (year)	41	29	28
Height (cm)	186	167	180
Weight (kg)	82	75	85
Head circumference (mm)[a]	599	597	583
C4 circumference (mm)	389	390	389
Neck length (mm)	140	133	131
h (mm)[b]	89	103	85
Frankfort plane angle (deg)[c]	19	17	20
Head CoG position (mm)[d]	(42,−136)	(23,−127)	(45,−129)
AO joint position (mm)[d]	(18,−99)	(−1,−90)	(21,−92)

[a]Maximum circumference, nominally parallel to Frankfort plane.
[b]Estimated vertical distance between the AO joint and line of action of the force applied during the MVC and the isometric sweep contractions (see Fig. 3(*a*)).
[c]While seated on sled.
[d](x,z) position relative to T1 centroid while seated on the sled.

flexor and extensor groups and then use similar activation intensities for all muscles within a group. Only a few models have varied individual muscle activation levels [11,21].

Validation of these complex neck models is difficult. Typical validation procedures consist of applying an input pulse (often at T1) and then comparing the head kinematics to human subject or cadaver data. Muscle activation levels are rarely based on in vivo recordings, but instead are tuned either to improve the fit between the model and validation data, or to meet some objective cost function. Neither of these techniques validates the muscle activation pattern used. The goal of this descriptive study is to generate a set of muscle and kinematic data for the validation of numerical models of the neck and its musculature. Three validation levels of increasing neuromuscular and kinematic complexity are considered: isometric (muscles active, but no motion), voluntary head movements (muscle activity generates movement) and impact conditions (externally induced kinematics that are modulated by reflex muscle activation). Kinematic data of the head and torso and electromyographic data of two anterior and six posterior neck muscles are reported.

Methods

Subjects. Three male subjects with no history of cervical injury, sensorimotor dysfunction or neck/back pain participated in the experiment (Table 1). Subjects gave their informed consent and were paid a nominal amount. The use of human subjects was approved by the university's Clinical Research Ethics Board and conformed to the Declaration of Helsinki.

Imaging. All subjects underwent multiple magnetic resonance (MR) scans to screen for existing pathology, to document neck muscle geometry, and to identify deep vasculature to avoid during wire insertions (Fig. 1). Scans were performed with a Phillips Gyroscan Intera 3.0 T magnet using 3 mm slices and a 0.3 mm spacing. Axial T1-weighted scans, proton density (PD) scans and T1-weighted fast field echo (FFE) scans were performed between the T3 spinal level and the occiput of each subject (Fig. 1).

Instrumentation. Electromyographic (EMG) activity was measured using a combination of indwelling and surface electrodes. Indwelling electrodes consisted of twisted pairs of insulated 0.05 mm wire (Stablohm 800A, California Fine Wire, Grover Beach, CA) with 0.5 mm of exposed wire at each recording tip. The ends of both wires in each twisted pair were hooked to anchor the wires in the muscle tissue. Wires were inserted into the left sternocleidomastoid (SCM), trapezius (Trap), levator scapulae (LS), splenius capitis (SPL), semispinalis capitis (SsCap), semispinalis cervicis (SsCerv), and multifidus (Mult) muscles using 25 gauge needles under ultrasound guidance (Sonos 5500, Agilent Technologies, Andover, MA). An anatomical description of these muscles is given in Table 2. All wire insertions were placed at the

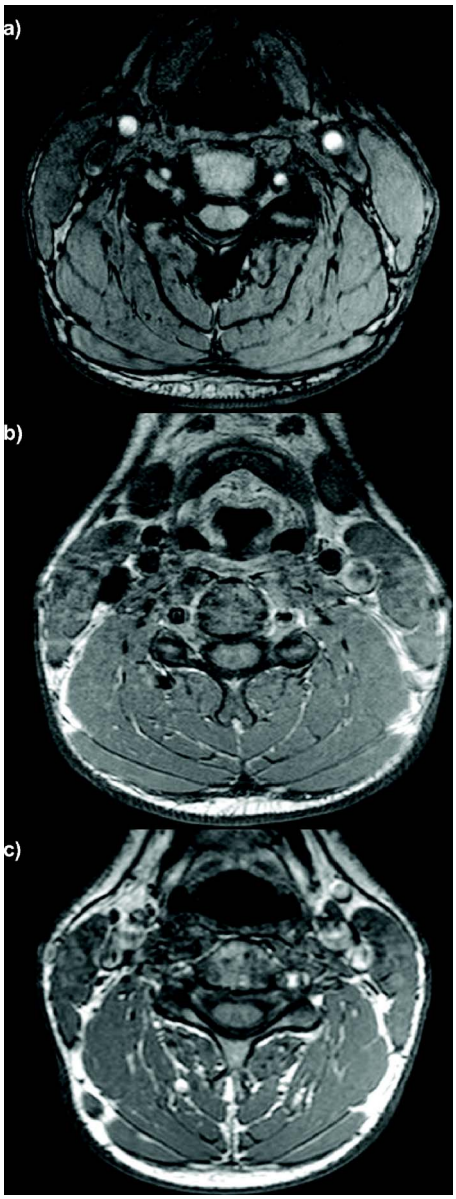

Fig. 1 Axial MR images at the C4 level of the three subjects. Shown are: (*a*) a fast field echo (FFE) scan of Subject 1; (*b*) a proton density (PD) scan of Subject 2; and (*c*) a T1-weighted scan of Subject 3. All three types of scans were acquired from each subject and all scans were acquired in the supine position.

C4/C5 level with an additional multifidus insertion at the C6/C7 level (Fig. 2). All wires were placed near the center of the muscle's horizontal cross section. In the SCM, the wire always remained superficial to the readily identifiable cleidomastoid subvolume [27]. During wire extraction, ultrasound was again used to confirm the wire had not moved. Although the wires were harder to visualize without the needle, the tissue displacement at the hooked end was readily apparent during wire extraction. Surface electrodes (H69P, Kendall-LTP, Huntington Beach, CA) were placed over the left sternohyoid (STH) muscle with 10 mm interelectrode spacing and a single surface electrode over the manubrium served as the reference electrode. Wire signals were amplified and band-pass filtered (30–1000 Hz) using a Neurolog system (Digitimer, Welwyn Garden City, Hertfordshire, UK) and surface signals were amplified and band-pass filtered (10–1000 Hz) using a Myosystem 1400 (Noraxon, Scottsdale,

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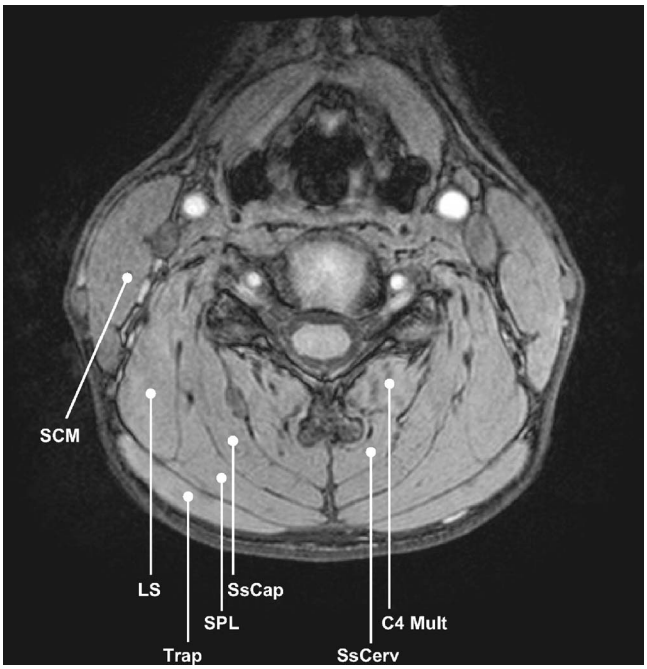

Fig. 2 Location of wire insertions into the neck muscles. Insertions shown on right side for clarity only; all wires inserted only on left side.

AZ) before being acquired at 2 kHz.

Isometric neck loads and moments were measured using a six-axis load cell (Bertec 4060H, ±10 kN vertical, ±5 kN horizontal, Worthington, OH; Fig. 3(*a*)). During dynamic tests, head acceleration was measured with a nine accelerometer array (Kistler 8302B20S1; ±20 g, Amherst, NY) arranged in a 3-2-2-2 configuration [28]. The accelerometers were mounted to a rigid aluminium frame that was securely strapped to a subject's head (Fig. 3(*b*)). Torso acceleration was measured with a triaxial linear accelerometer (Summit 34103A; ±7.5 g, Akron, OH) and angular rate sensor (DynaCube; ±100 rad/s, ATA-Sensors, Albuquerque, NM). Both torso transducers were fastened to a plate secured immediately below the sternal notch with adhesive and straps. Sled acceleration was measured with a uniaxial accelerometer (Sensotec JTF3629-05; ±20 g, Columbus, OH) oriented horizontally along the axis of motion.

Displacements were measured with a motion analysis system (Pheonix VZ4000, Burnaby, BC) and nine markers placed on the head, torso, and sled. Transducer signals were low-pass filtered (1000 Hz) and then simultaneously sampled at 2 kHz. Displacement data were acquired at 100 Hz per marker. The location and orientation of the head and torso instrumentation was measured relative to anatomical landmarks using a three-dimensional (3D) digitizer (FaroArm B08-02, single-point accuracy ±0.30 mm, Lake Mary, FL).

Test Procedures. Once instrumented, seated subjects performed four tasks: (i) maximal voluntary contractions (MVC) in eight fixed directions (45 deg intervals); (ii) 50 N isometric neck muscle contractions with a sweeping force direction; (iii) voluntary oscillatory head movements; and (iv) reflex muscle activations to a forward (+*x*) acceleration. For the MVCs and isometric sweep tasks, subjects had their head firmly clamped to an overhead load cell and their torso firmly strapped to a rigid seat back (Fig. 3(*a*)). This arrangement allowed subjects to apply a force in any direction within a horizontal plane located at the height of their forehead and opisthocranion without moving their head. MVCs were held for 5 s in each direction and separated by rest periods of 30–60 s. The eight different MVC directions were presented in an order that minimized fatigue. Subjects received verbal encouragement and visual feedback of their applied force direction and magnitude on a real-time 2D plot of their anteroposterior (AP) and mediolateral (ML) reaction forces. Subjects practiced MVCs in some but not all directions prior to testing to familiarize themselves with the task and the visual feedback. MVC trials in which the force angle was not maintained within 5 deg of the target angle were repeated at the end of the block of eight trials.

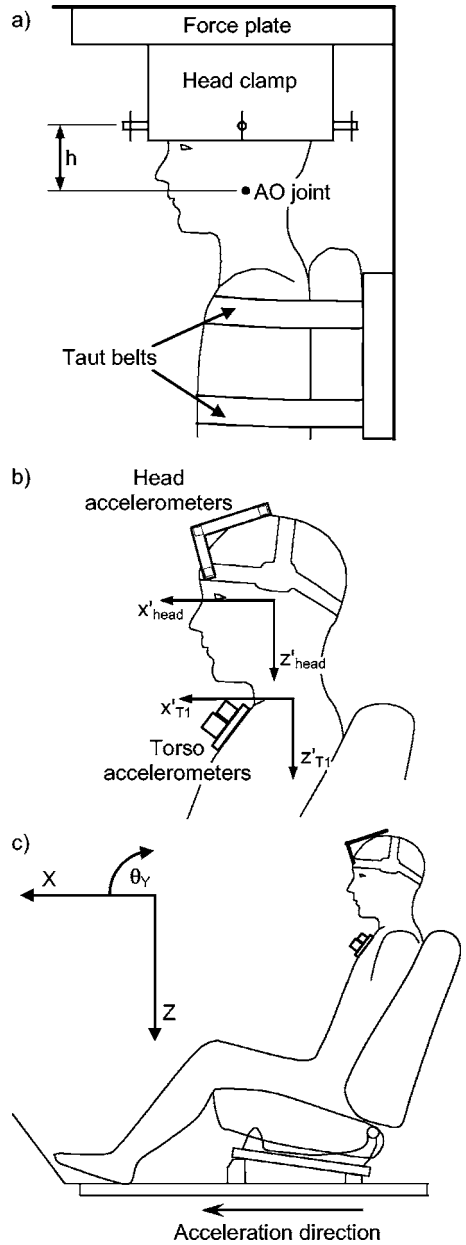

Fig. 3 Schematic showing: (*a*) the head clamp and force plate used for the isometric contractions; (*b*) the transducers and reference frames used for the voluntary movements and sled tests; and (*c*) the sled configuration.

After a rest period, subjects practiced applying a 50 N horizontal force against the load cell and slowly sweeping the direction of the applied force in the horizontal plane through 720 deg (two complete sweeps, beginning with extension). A fixed, submaximal force level of 50 N was chosen to keep the task simple, to avoid the fatigue that would develop during a maximal effort for the 20 s needed to complete this task, and to eliminate the need for

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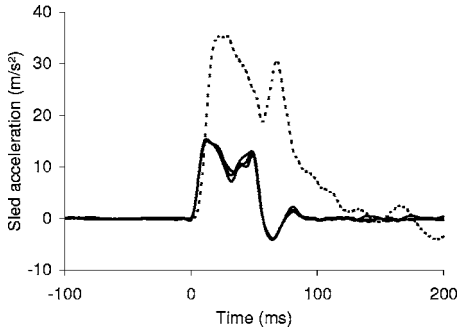

Fig. 4 Superposition of the sled acceleration pulses for the three subjects (solid lines) and a vehicle-to-vehicle collision pulse (dashed line) with a speed change of 8 km/h recorded during earlier experiments [48].

subjects to modulate both force amplitude and direction had we chosen to using a fixed percentage of a subject's maximal force. A 50 N circle was superimposed on the 2D visual-feedback plot of AP and ML forces to assist subjects in maintaining 50 N throughout the entire sweep. No head motion occurred during this task, but the direction of the force generated by the neck muscles and applied by the head to the overhead load cell changed continuously throughout the task. Once able to perform the task smoothly, subjects performed four isometric sweep contractions, two each in the clockwise ($+z$) and counterclockwise ($-z$) directions, over a period of about 20 s per trial. The middle 360 deg (about 10 s of data) of each trial was subsequently used for analysis.

For the voluntary and reflex-activation tasks, participants were seated in a 1991 Honda Accord car seat with the head restraint removed (Fig. 3(c)). The seat was mounted on a custom-fabricated sled powered by feedback-controlled linear induction motors (Kollmorgen IC55-100A7, Kommack, NY). The sled generated no audible or mechanical preperturbation signals that participants could use to predict perturbation onset. Subjects were instructed to adopt a comfortable seated posture, face forward, rest their forearms on their lap, and relax their face and neck muscles. To eliminate habituation effects [29,30], subjects received a single forward horizontal perturbation ($a_{peak}=1.55\pm0.03$ g; $t_{peak}=16.6\pm0.3$ ms; $\Delta t=59.0\pm0.3$ ms; $\Delta v=0.499\pm0.003$ m/s, Fig. 4). After acceleration, the sled continued at constant speed for 0.5 s before decelerating to rest. After the perturbation and while still seated on the sled, subjects performed two voluntarily oscillatory flexion/extension neck/head movements. For each movement, subjects started from the neutral rest position and cycled through a comfortable range of flexion and extension movements five times at 1 Hz (a metronome provided a timing signal).

Data Reduction. All wire EMG data were high-pass filtered (50 Hz) to remove motion artifact before calculating the root-mean-squared (rms) values using a moving 20 ms window. For each MVC trial, a moving average was used to find the 500 ms interval of maximal force and the maximum rms EMG for each muscle was then computed for the same interval. To generate a normalized tuning curve for each muscle, the rms values from each of the eight directions were normalized by a muscle's maximum rms value. For the isometric sweep contractions, the force and normalized rms EMG were averaged over a moving 5 deg interval to generate continuous tuning curves. Moments about the atlanto-occipital (AO) joint for the MVCs were computed by multiplying the resultant horizontal force by the moment arm (h) between the AO joint (estimated from the height of the mastoid process) and the line of action of the force measured by the load cell (Fig. 3(a), Table 1).

For the voluntary movements and sled perturbations, rigid body transformations were used to resolve the head and torso transducers into two global reference frames with origins at the head center of mass and center of the T1 vertebral body, respectively. The x axis was horizontal and positive forward, the z axis was vertical and positive down, and the y axis was horizontal and positive to the right (neck extension was positive about the y axis). The head center of mass was assumed to be directly above (orthogonal to the Frankfort plane) the midpoint of the interaural axis by a distance equal to 17% of the vertical distance between the interaural axis and the vertex [31] (Table 1). Based on 41 subjects from a previous study [32], this CoG position is 23 mm±7 mm forward and 6 mm±5 mm below the position predicted by the regression equations published by Clauser et al. [33]. Based on MR data from the same 41 subjects, the AO joint in the current subjects was assumed to be 24 mm behind and 37 mm below the head center of mass (Table 1). The center of the T1 vertebral body relative to the manubrium and C7 spinous process was determined from each subject's MR scan. Both external landmarks (manubrium and C7 spinous process) were then digitized with the subject seated on the sled.

All accelerations were determined from the transducers and all displacements were determined from the motion analysis system. All linear accelerometers were corrected for the time-varying orientation of the earth's gravity field prior to computing the six degree-of-freedom kinematics. Peak angular velocity at T1 was determined from the angular rate sensor data after it had been digitally compensated to reduce the sensor's high-pass frequency to 0.002 Hz [34]. T1 angular accelerations were then computed by finite differences (5 ms window) from the compensated angular velocity data. The kinematics of the head relative to T1 were computed by subtracting the time-varying T1 kinematics from the time-varying head kinematics in the lab frame. The angular velocity of the head with respect to T1 was computed by integrating the relative angular acceleration. Retraction was defined as the horizontal rearward translation of the AO joint relative to the moving T1 origin. A subject's initial head and torso position and orientation were determined from the motion analysis markers over a 100 ms period preceding the perturbation and all time-varying kinematic signals were set to zero at the start of the movement or perturbation.

Results

The peak horizontal forces generated during the MVCs varied from a minimum of 118 N in anterolateral flexion to a maximum of 323 N in extension (both in Subject 3, Fig. 5). These forces equated to moments about the AO joint of 10 and 29 N m respectively. All nine muscles were active over a well-defined directional range between 90 deg and 180 deg wide; outside of this range, a muscle was essentially quiet (see bottom four rows, Fig. 5). The preferred activation directions for STH (second row, Fig. 5) and both multifidus muscles (bottom row, Fig. 5) were bilaterally symmetrical, whereas the preferred activation directions for the other six muscles were biased toward the ipsilateral side. Except for SPL, the preferred activation directions of all muscles were similar in all three subjects and consistent with their anatomical position relative to the spine. SPL however, despite its posterior anatomical location, was primarily active during lateral and anterolateral flexion, and on average more than 50% active during pure flexion (dashed line, fourth row, Fig. 5). In Subject 2, SPL remained more than 50% active during anterolateral flexion toward the contralateral side—a direction opposite to its anatomical position in relation to the spine.

The 50 N isometric sweep contractions generated muscle tuning curves that were similar (but not identical) to those generated by the MVCs (compare Fig. 5 to Figs. 6 and 7). Some muscles, particularly SCM in Subjects 1 and 3 (third row, Fig. 6) and semispinalis capitis in Subject 2 (second row, Fig. 7), remained active throughout the 360 deg sweep, whereas other muscles, particularly trapezius (bottom row, Fig. 6), were active over a larger angular range during the sweep than during the MVCs. All three

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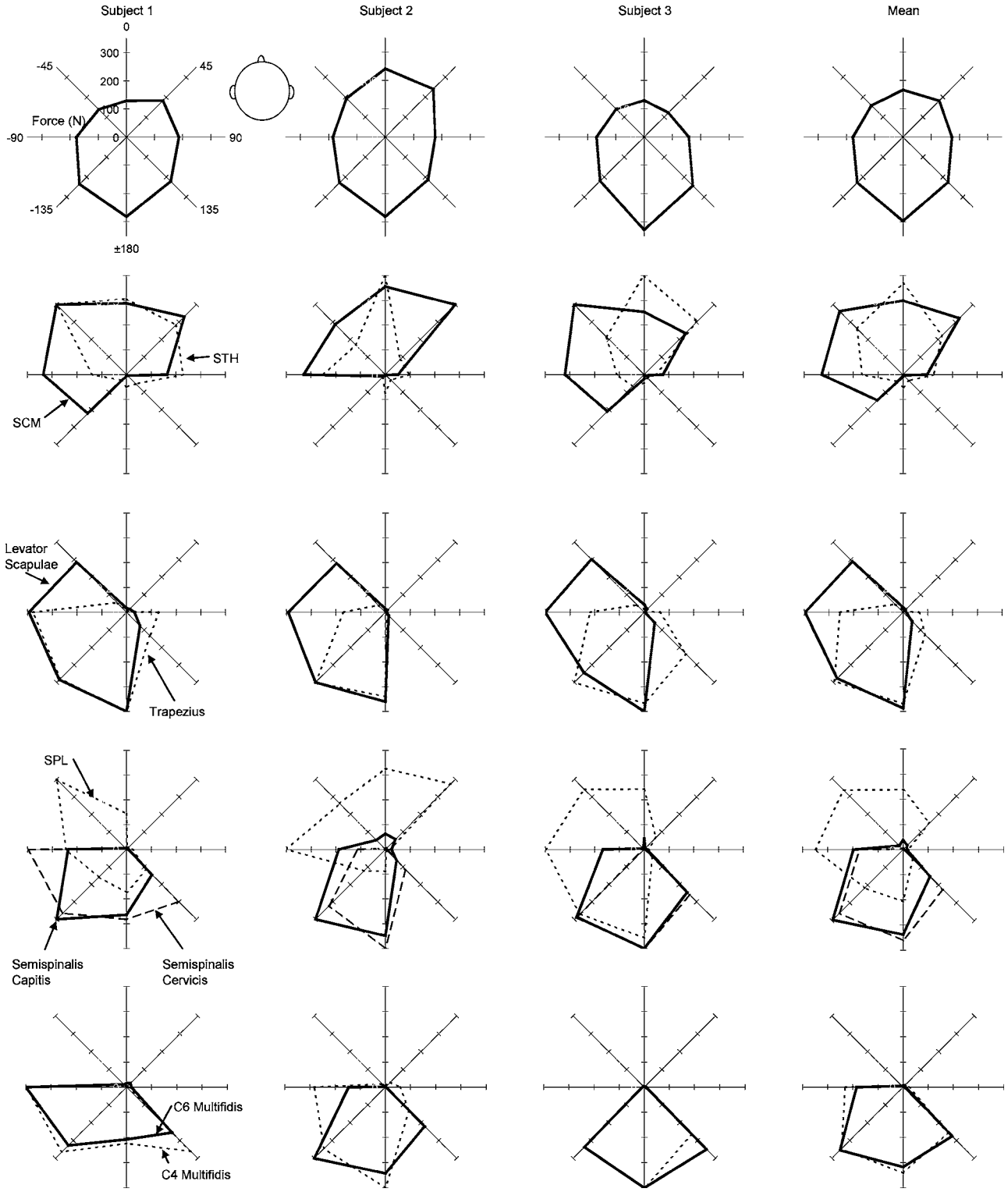

Fig. 5 Horizontal force (top row) and directional muscle tuning curves (bottom four rows) for each subject (left three columns), and the mean response of all subjects (right column). Scale and direction shown at top left. All muscle activation levels normalized to the maximum value (100%) for that particular subject's muscle.

subjects exhibited similar tuning curves for STH (second row, Fig. 6), levator scapulae (fourth row, Fig. 6) and C6 multifidus (bottom row, Fig. 7). Except for SPL, the tuning curves of the remaining muscles were similar for Subjects 1 and 3, but smaller for Subject 2. Aside from two isolated bursts of activity, SPL remained relatively quiet in Subjects 1 and 2, but relatively active in Subject 3 (first row, Fig. 7).

The amplitude of the muscle tuning curves did not vary with sweep direction (compare thick and thin lines in Figs. 6 and 7), but some of the clockwise and counterclockwise tuning curves were slightly rotated with respect to one another. Generally, the angle at which a muscle became active during a sweep in one direction occurred slightly before the angle at which it became inactive in the opposite direction. One exception to this general

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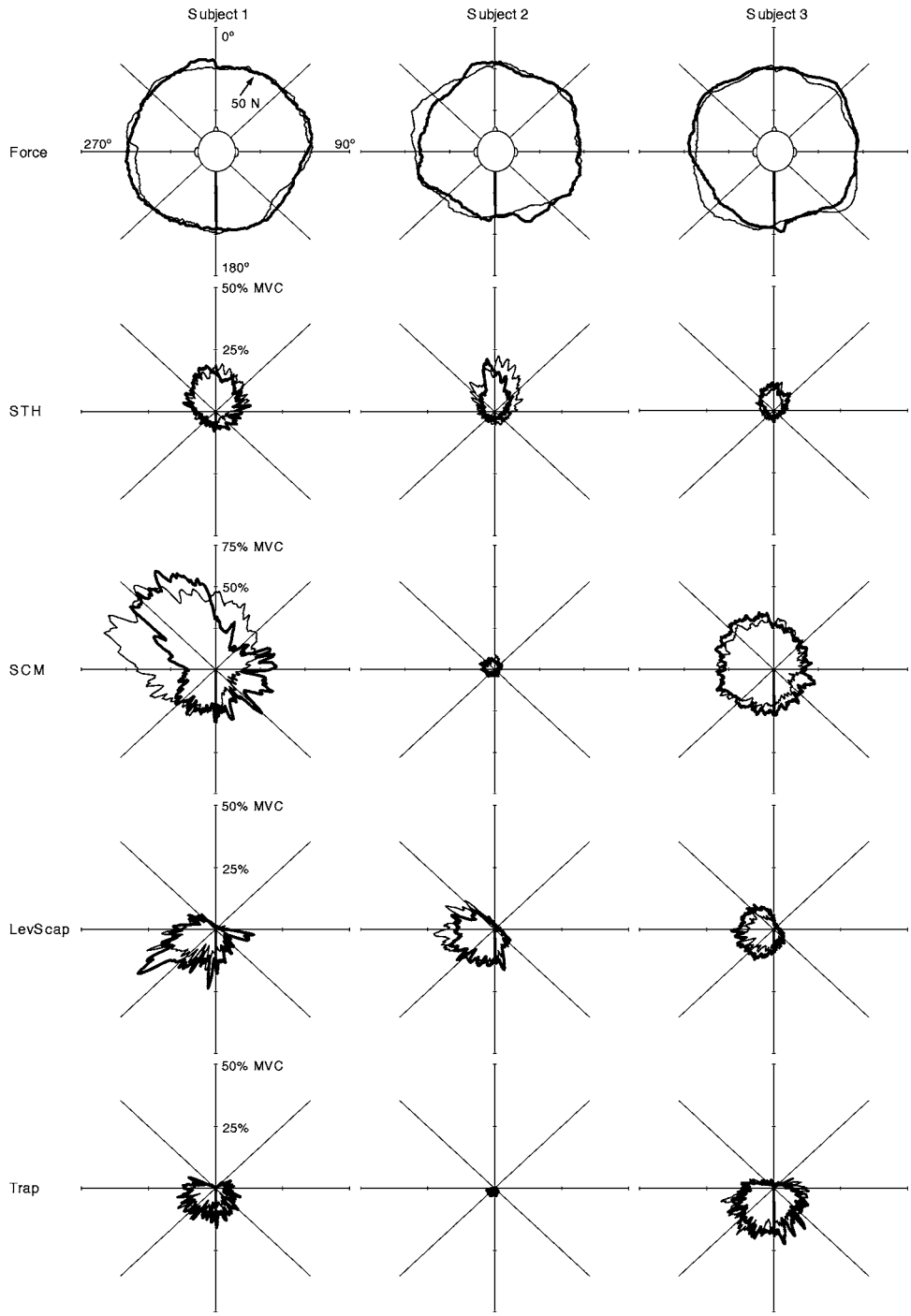

Fig. 6 Horizontal force (top row) and tuning curves (bottom four rows) for sternohyoid (STH), sternocleidomastoid (SCM), levator scapulae (Lev Scap), and trapezius (Trap) during the 50 N isometric force sweeps. The dashed circle in the force plots (top row) indicates 50 N. Clockwise (+*z*) sweeps shown with a thick line and counterclockwise (−*z*) sweeps shown with a thin line.

observation was SCM in Subject 1 (third row, Fig. 6), which exhibited elevated activation levels over a wider angle during the counterclockwise sweep than during the clockwise sweep.

Voluntary flexion/extension movements were generated using a similar strategy in all three subjects (Fig. 8). STH and SCM muscles acted synergistically to accelerate the head into flexion ($-\alpha$, between dashed lines *A* and *B* in Fig. 8), whereas semispinalis capitis, semispinalis cervicis, and both multifidus muscles acted synergistically to accelerate the head into extension ($+\alpha$, between dashed lines *B* and *C* in Fig. 8). Levator scapulae and trapezius activity was less consistent—perhaps due to their anatomical association with the shoulder—but levator scapulae did exhibit a large burst during the onset of the first oscillation in Subjects 1 and 2 (third row, Fig. 8). SPL again varied between subjects (fifth row, Fig. 8): it was silent in Subject 1, exhibited a variable pattern in Subject 2, and was generally in phase with the posterior muscles in Subject 3.

The forward sled perturbation produced similar early head and torso kinematics in all three subjects (Fig. 9). Subject motion appeared first as forward ($+x$) and upward ($-z$) T1 accelerations and

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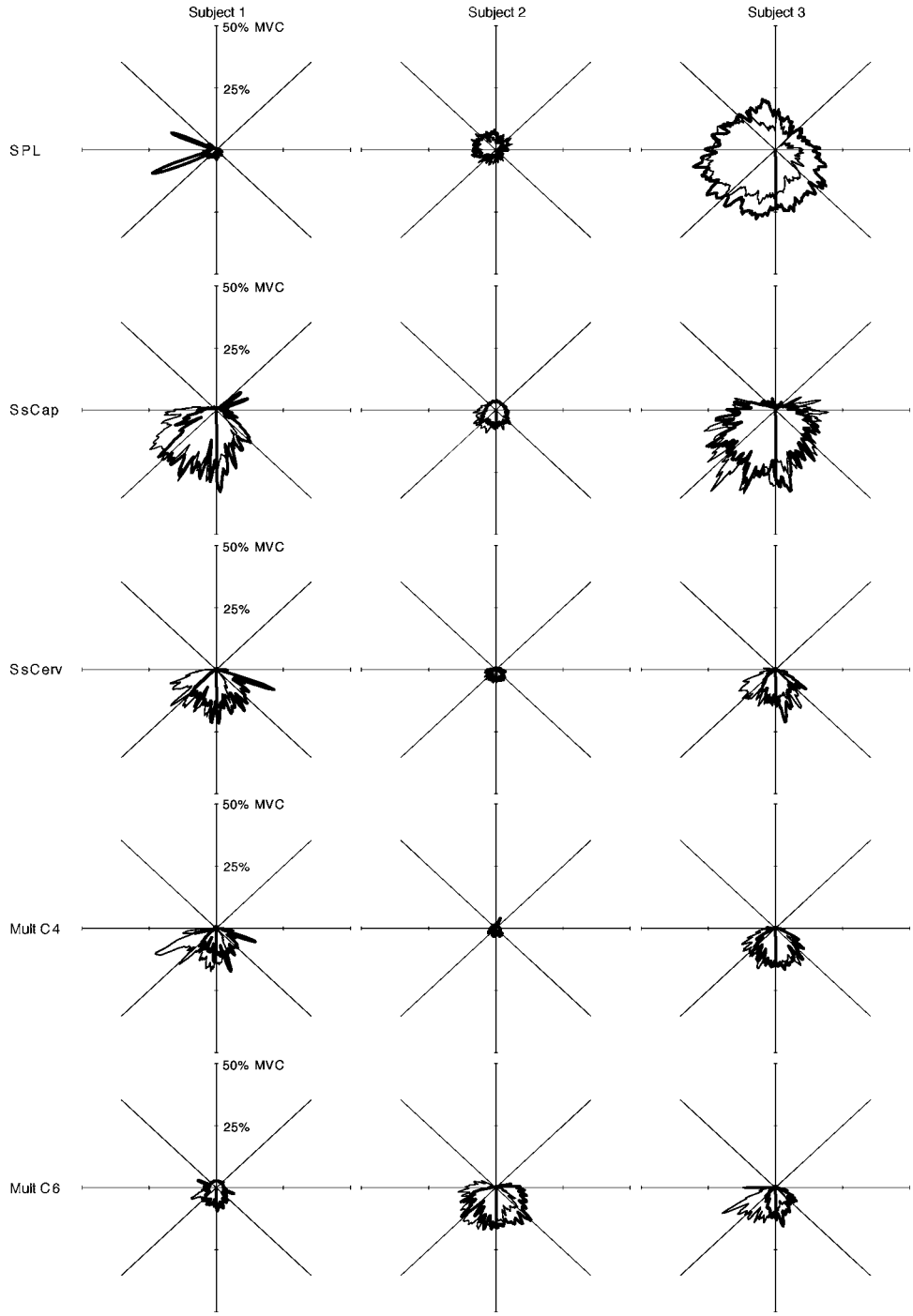

Fig. 7 Tuning curves for splenius (SPL), semispinalis capitis (SsCap), semispinalis cervicis (SsCerv), and multifidus at the C4 (MultC4) and C6 (MultC6) levels during the 50 N isometric sweeps (continuation of Fig. 6). Clockwise (+*z*) sweeps shown with a thick line, and counterclockwise (−*z*) sweeps shown with a thin line.

were followed by linear (+x) and angular (+y) accelerations of the head. This lag between T1 and the head produced both neck retraction and combined head/neck extension (θ) (Fig. 9). All three subjects had similarly early STH and SCM activation, with peak levels between 0.5 to 1.9 times MVC. This anterior muscle activation preceded peak forward linear acceleration (a_x) of the head with respect to T1 (see dashed vertical lines in Fig. 9) and returned to baseline by the time peak head extension occurred about 190 ms after perturbation onset. Posterior muscle activity was also present during the early period of STH and SCM activity, although it was not consistent across subjects. In Subject 1, levator scapulae, semispinalis capitis, and both multifidus muscles were active at the same time as STH and SCM, with both levator scapulae and C6 multifidus approaching 50% MVC activation levels. In contrast, both Subjects 2 and 3 exhibited early activity in SPL, which reached 65% MVC levels in Subject 2.

During the rebound motion phase, all three subjects rapidly restored their upright head posture; however Subject 3 did so without eliminating the retraction that had also developed (see Retraction in the third column of Fig. 9). There was generally

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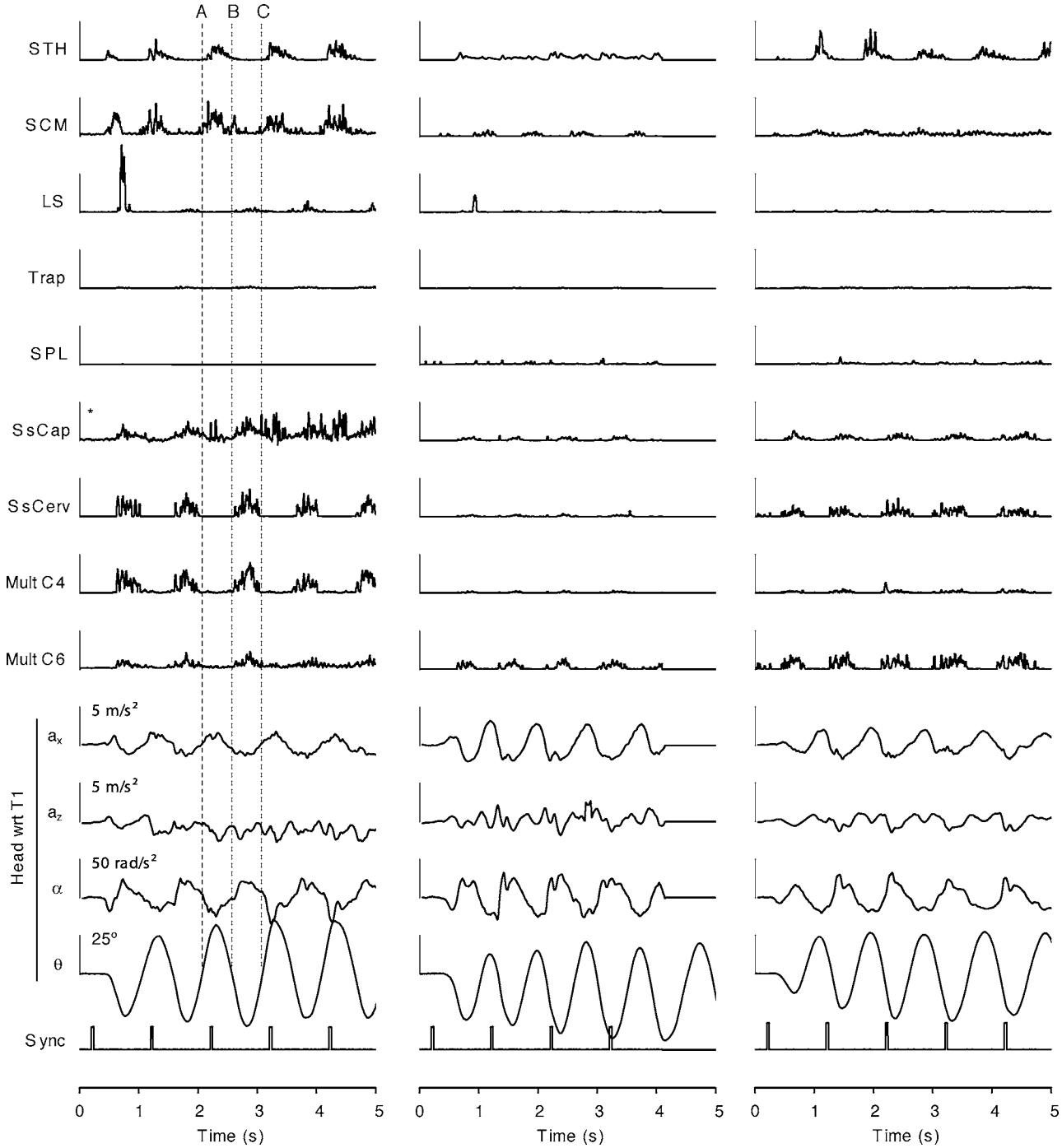

Fig. 8 Normalized rms electromyographic recordings (scale bar=100% MVC) and kinematics of the head with respect to T1 for voluntary head flexion/extension movements executed by subjects 1 (left), 2 (middle), and 3 (right). Sync refers to an acoustic signal heard by the subjects. The head is extended and being accelerated in the flexion direction during time interval AB between the left two dashed vertical lines, and the head is flexed and being accelerated in the extension direction during time interval BC (* increase in baseline noise for SsCap in Subject 1 after sled test but before flexion/extension movement suggests the normalization may be invalid). Use only relative amplitude and timing from this trace): (*a*) linear acceleration: (α) angular acceleration; (θ) angle.

less muscle activity during the rebound phase than the earlier retraction phase. Semispinalis capitis activity was observed in all three subjects during rebound, with Subject 2 also enlisting his deeper semispinalis cervicis and multifidus muscles at both submaximal and supramaximal levels (middle column, Fig. 9).

Discussion

The activation patterns observed during the isometric, voluntary, and reflex contractions used in this study were consistent for some muscles but variable for other muscles. The two anterior muscles—the sternohyoid muscle and the cleidomastoid subvol-

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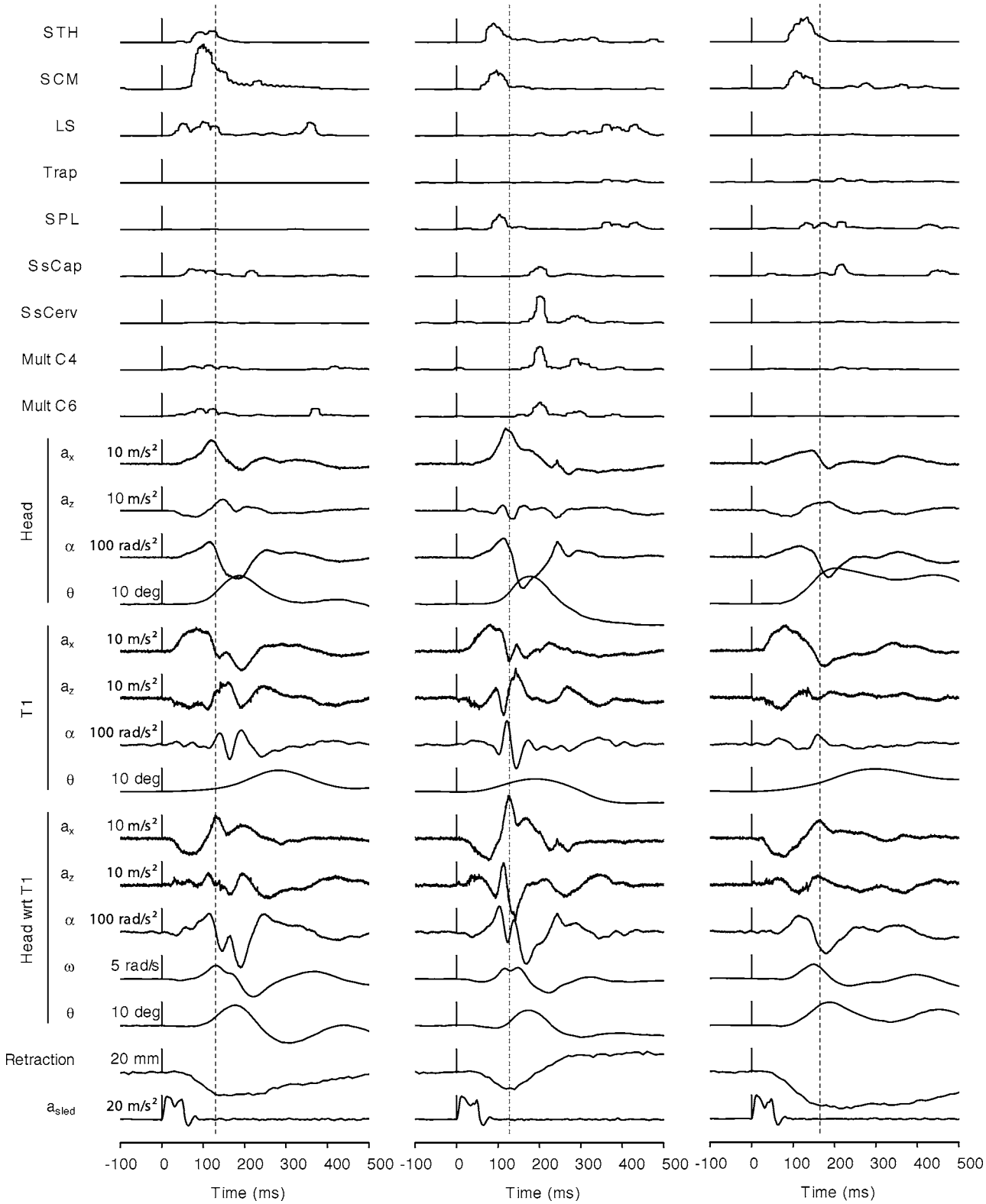

Fig. 9 Normalized rms EMG and kinematics of the head, T1 and head with respect to T1 for forward sled tests experienced by subjects 1 (left), 2 (middle), and 3 (right). Dashed vertical lines aligned with peak horizontal acceleration (ax) of the head with respect to T1. Muscle scale bars indicate 100% of maximal voluntary contractions (MVCs).

ume of the sternocleidomastoid muscle—had similar preferred activation directions that were consistent with their anatomical locations. They behaved similarly during both isometric tasks and acted synergistically in both dynamic tasks. With the exception of SPL, the posterior and posterolateral neck muscles also showed activation directions consistent with their anatomical locations and acted synergistically during the voluntary motions. They did not, however, exhibit similar activation strategies across subjects during the sled perturbation.

Of the posterior neck muscles studied, SPL exhibited the most

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Table 2 Action, Origin, Insertion, and Innervation of the Instrumented muscle[a]

Muscle	Action	Origin	Insertion	Innervation
Sternohyoid	Depresses the hyoid bone	Posterior surface of the medial end of the clavical, the posterior sternoclavicular ligament and the upper posterior aspect of the manubrium sterni	Inferior border of the body of the hyoid bone	Branches of the ansa cervicalis (C1, 2, 3)
Sternocleidomastoid	Rotates and flexes neck	By two heads, from sternum and clavicle	Mastoid process and lateral half of the superior nuchal line	Accessory nerve and possibly branches of the ventral rami of C2, C3, and sometimes C4 spinal nerves
Levator scapulae	Elevates posterior angle of scapula	Transverse processes of C1, 2, 3, and 4	Superior edge of scapula.	C5 via the dorsal scapular nerve and branches of the C3 and C4 spinal nerves
Trapezius (superior third)	Extends and laterally flexes the head (with shoulder fixed) and elevates shoulder (with head fixed)	Medial third of the superior nuchal line, external occipital protuberance, ligamentum nuchae and spinous processes of the cervical (and thoracic) vertebrae	Lateral third of the clavicle with lower fibres wrapping on to the medial margin of the acromium and scapular spine	Accessory nerve
Splenius	Extends and rotates the neck	Spines of the C7–T4 vertebrae and dorsal edge of the lower half of the ligamentum nuchae	Mastoid and lateral third of the superior nuchal line (deep to Sternocleidomastoid)	Lateral branches of the dorsal rami of the lower cervical spinal nerves
Semispinalis capitis	Rotates and draws head backward.	Transverse processes of T7–C7 and articular processes of C4–C6.	Medial part of the occipital bone between the inferior and superior nuchal line.	Branches of cervical nerves.
Semispinalis cervicis	Extend and rotate the cervical and thoracic spine	Transverse processes of upper five or six thoracic vertebrae	Spinous process of C2–C5	Dorsal rami of the cervical and thoracic spinal nerves
Multifidus	Rotates spinal column.	Sacrum, iliac spine, lumbar, cervical, and dorsal vertebrae.	Laminae and spinous processes of next four vertebrae above.	Dorsal rami of the nerves.

[a]Source: Gray's Anatomy, 38th ed., 1995, Churchill Livingstone, New York, and *Taber's Cyclopedia Medical Dictionary* 18th ed., 1997, FA Davis Company, Philadelphia, PA.

varied behavior. Despite accounting for 30% of the total extensor moment generating potential of the human neck [20], SPL was most active during lateral and anterolateral flexion and, in one subject, active even during anterolateral flexion toward the contralateral side. Previous intramuscular recordings of SPL under isometric conditions have shown two directional patterns: a consistent posterolateral directional preference [35,36] or a subject-specific anterolateral and/or posterolateral directional preference [37,38]. Moreover, SPL has a relatively broad insertion along the superior nuchal line, with fibers that insert near the mastoid acting as antagonists with the ipsilateral SCM and synergists with the contralateral SCM to generate or control axial head rotation [39]. Thus studies, like ours, that observed both anterolateral and posterolateral direction preferences in SPL may have recorded from more lateral fibers than studies that observed a more consistent posterolateral direction preference in SPL. Since none of the previous studies reported the position of their electrodes within this relatively large muscle, further work is needed, perhaps with multiple wire insertions, to better understand the variable roles (and possibly different compartments) of the splenius muscle.

Despite the intersubject and interstudy differences related to SPL, this muscle exhibited a consistent directional preference within each subject studied here. In the subject who used SPL during maximal isometric flexion rather than extension (Subject 2), SPL was active with STH and SCM when a flexor moment was needed to resist extension during the early part of the sled perturbation. In the subject who used SPL during both maximal isometric flexion and extension contractions (Subject 3), SPL was active in both the late retraction and early rebound phases of head motion during the sled perturbation. This within-subject consistency in SPL behavior supports the earlier proposition that SPL may have multiple, functionally different compartments.

The consistent directional behavior of SPL was not exhibited by all posterior neck muscles studied here. The C4 and C6 multifidus muscles had similar directional tuning curves in all subjects and were active during angular acceleration in the extension direction ($+y$) in the voluntary flexion/extension movements. During the sled perturbations, however, the multifidus muscles were active during the initial retraction/extension phase in one subject, during the rebound phase in another subject, and during neither phase in the third subject. The reason for this variable reflex response in multifidus is not clear, but could be related to subject experience: all three subjects had previously experienced the perturbation (>1 month prior to testing) and knew it was relatively benign. This knowledge may have eliminated an element of surprise [40] and reduced the likelihood of evoking a protective co-contraction of the anterior and posterior muscles [29,30]. More work is needed in inexperienced subjects to better understand this issue.

Some of the muscle tuning curves exhibited isolated spikes of EMG activity (see in particular SPL for Subject 1, Fig. 7). The raw EMG for these trials showed large isolated action potentials (both individually and in trains) and suggested that the 50 N isometric sweep task was at the threshold for recruiting these motor units. Similar reasoning may explain the low EMG levels for Subject 2 during the isometric sweep and flexion/extension tasks (Figs. 6–8). The presence of large, clear EMG activity both before (MVC) and after (sled perturbation) these tasks, combined with similar baseline noise in all tasks, indicates that deterioration of this subject's EMG was not responsible for the low EMG levels. More likely, the isometric sweep and flexion/extension tasks were not demanding enough to recruit the large motor neurons enlisted by Subject 2 during his MVC and sled perturbation tasks. Subject 2's lower isometric sweep force in the posterior quadrant (average of 40 N between 135 deg and 225 deg, Fig. 6) may have also contributed to the lower posterior muscle recruitment during the isometric sweep task. Similarly, Subject 2's greater flexor neck strength (top row, Fig. 5) and lower amplitude kinematics (Fig. 8) could have contributed to the lower EMG levels seen in this subject's anterior and posterior muscles during the voluntary flexion/extension task.

The neuromuscular and kinematic data presented here have a number of implications for numerical modeling of the human neck. First, the synergy observed between STH and SCM, combined with previous work showing a similar synergy between SCM and the longus colli [41], suggests that the common modeling assumption of uniform activation of the anterior neck muscles

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appears to be reasonable. Second, the large and consistent activity in STH, combined with a flexor moment arm considerably larger than other neck flexors (e.g., SCM and longus colli), suggests that it should be included in neck muscle models examining sagittal plane motion. And third, the assumption of uniform activation of the posterior neck muscles appears to be incorrect. SPL in particular has a unique activation profile and directional preference, and given its large physiological cross-sectional area and a correspondingly large moment arm [20] likely needs activation dynamics modeled independently of the other posterior muscles.

Numerical neck models are also based on an assumption that muscle recruitment patterns are similar between individuals. While the current data indicate that this assumption is reasonable both for the anterior muscles during the four tasks studied here and the posterior muscles (save SPL) during the isometric and voluntary tasks, it does not appear to hold for the posterior muscles in the sled perturbation and in particular the retraction phase in which whiplash injury may occur. Although multiple strategies for posterior neck muscle recruitment are perhaps not surprising in a system as complex and redundant as the human neck, this finding suggests that modelers should not validate their models by first optimizing muscle activation levels based on some objective cost function and then comparing their optimized muscle activation levels to in vivo recordings of neck muscle activity. Instead it suggests that neck models need to be validated using in vivo muscle recordings—possibly from a number of subjects using different recruitment patterns—as inputs, and then a comparison made between the kinematic responses of the model and the human. This latter procedure would allow one to study the effect of specific posterior muscle activation patterns on the overall kinematic response and, in more complex models, the individual tissue responses.

During the dynamic tasks, a number of the muscles studied here exceeded the activation levels measured during the MVCs. Such supramaximal activity has been previously reported in the SCM by Kumar et al. [42], although the reason for this phenomenon is not clear. One possible explanation is that subjects simply did not exert maximal efforts during their MVCs; however, the neck moments measured in our subjects (minimum 10 N m in flexion and maximum 29 N m in extension) are in line with previously reported maximal neck moments (resolved to the mastoid) of 13 N m±3 N m in flexion and 24 N m±7 N m in extension [43].

Alternatively, it is possible that some muscles cannot be fully recruited during contractions designed to maximize the horizontal force applied by the head at the level of the forehead. For instance, SCM may require combined flexion and axial rotation to recruit the muscle maximally, although this explanation seems unsatisfactory given that SCM was recruited at supramaximal levels during the sagittal plane motion induced by the sled perturbation. Alternatively, short, reflex muscle activation generates more synchronous bursts of action potentials than produced during a 5 s maximal voluntary effort, and the summation of these synchronous action potentials yields a greater EMG signal. It seems unlikely, however, that this synchronization could explain the twofold increase in activation levels observed in the SCM of Subject 1. Whatever the reason, these supramaximal measurements mean that modelers cannot simply use the normalized EMG data to scale muscle force from a muscle's physiological cross-sectional area (PCSA). Instead, they may need to first develop individual muscle scaling parameters by modeling the MVC contractions, and then use these scaling parameters during subsequent dynamic modeling. Further work is needed to determine why these apparently supramaximal activation levels occur and how to best contend with them from a modeling perspective.

The current results also have implications for whiplash injury—particularly injury to the facet capsular ligaments. Recent studies have shown that the cervical multifidus muscles originate from the facet capsular ligaments of the C4-5, C5-6, and C6-7 facet joints [44,45]. Our data show that in some subjects these deep muscles are active during abrupt forward accelerations. Such reflexively-evoked contractions have the potential to increase the strain in the facet capsular ligament above that previously measured under passive loading conditions [46,47] and thereby increase the potential for injury to this structure. Moreover, the intersubject variability in multifidus activation levels—from almost no activity in one subject to near maximal activity in another subject—may also explain why whiplash injury does not develop in all individuals exposed to a particular impact.

In the current study, we limited our analysis to two dynamic conditions that generated head and neck motions in the sagittal plane. The neck can be moved or perturbed in other directions and further work is needed to document the neuromuscular behavior of the superficial and deep muscles under these different conditions. We were also limited to eight in-dwelling recordings despite 20 or more muscle pairs in the human neck [16]. The muscles we chose represent a large proportion of the muscle tissue at the C4 level, but the contribution of the muscles we did not measure and the contribution of other compartments within the muscles we did measure remain unknown. In addition, only male subjects were presented in the current study, and thus gender differences in neck muscle activation and recruitment pattern remain to be studied.

The speed change and peak acceleration of the collision pulse used in this study is lower than many rear-end collisions. A low severity pulse was used because the head restraint was removed to ensure the observed head kinematics were generated entirely by loads developed in or transmitted through the neck. With both a higher collision severity and a head restraint, peak head kinematics would be driven largely by head-to-head-restraint interaction—a phenomenon which would have reduced the utility of the current data for validating numerical neck models. Despite a lower speed change and peak acceleration, the initial 16 ms of the collision pulse is very similar to larger pulses (speed change =8 km/h) used in previous studies [32,48] (see dashed line in Fig. 4). Moreover, previous work using surface electromyography has demonstrated a dose–response relationship in muscle activation levels with impacts varying between 1.2 km/h and 8 km/h [15,49], although further work is needed to confirm whether this global pattern applies to individual posterior muscles.

In summary, the superficial and deep muscle of the human neck generally exhibited consistent directional tuning and activation during the different isometric, voluntary, and reflexive tasks used in this study. The anterior muscles were more consistent than the posterior muscles, and the splenius muscle in particular displayed the most variability between subjects and across tasks. Despite these differences, the descriptive data presented in this study can be used to validate numerical models of the neck musculature for tasks of increasing kinematic and neuromuscular complexity.

Data Availability

The authors will make available digital files of the muscle, kinematic and kinetic data to researchers developing or validating numerical neck models. Requests for the data should be directed to Dr. Gunter Siegmund at gunter.siegmund@meaforensic.com

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References

[1] Mertz, H. J., and Patrick, L. M., 1971, "Strength and Response of the Human Neck (710855)," *Proceedings 15th Stapp Car Crash Conference*, Coronado, CA, November 17–19, Society of Automotive Engineers, Warrendale PA, pp.

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207–255.
[2] Culver, C., Neathery, R., and Mertz, H., 1972, "Mechanical Necks With Humanlike Responses (720959)," *Proceedings 16th Stapp Car Crash Conference*, Detroit, MI, November 8–10, pp. 61–75.
[3] Wilke, H. J., Claes, L., Schmitt, H., and Wolf, S., 1994, "A Universal Spine Tester for In Vitro Experiments With Muscle Force Simulation," Eur. Spine J., **3**(2), pp. 91–97.
[4] Bernhardt, P., Wilke, H. J., Wenger, K. H., Jungkunz, B., Bohm, A., and Claes, L. E., 1999, "Multiple Muscle Force Simulation in Axial Rotation of the Cervical Spine," Clin. Biomech. (Bristol, Avon), **14**(1), pp. 32–40.
[5] Ivancic, P. C., Panjabi, M. M., Ito, S., Cripton, P. A., and Wang, J. L., 2005, "Biofidelic Whole Cervical Spine Model With Muscle Force Replication for Whiplash Simulation," Eur. Spine J., **14**, pp. 346–355.
[6] Pontius, U. R., and Liu, Y. K., 1976, *Neuromuscular Cervical Spine Model for Whiplash (760770)*, Society of Automotive Engineers, Warrendale, PA.
[7] Williams, J. L., and Belytschko, T. B., 1983, "A Three-Dimensional Model of the Human Cervical Spine for Impact Simulation," J. Biomech. Eng., **105**(4), pp. 321–331.
[8] de Jager, M., Sauren, A., Thunnissen, J., and Wisman, J., 1996, "A Global and a Detailed Mathematical Model for Head-Neck Dynamics (962430)," *Proceedings 40th Stapp Car Crash Conference*, Albuquerque, NM, November 4–6, Society of Automotive Engineers, Warrendale PA.
[9] van der Horst, M. J., Thunnissen, J. G. M., Happee, R., van Haaster, R. M. H. P., and Wisman, J. S. H. M., 1997. "The Influence of Muscle Activity on Head-Neck Response During Impact (973346)," *Proceedings 41st Stapp Car Crash Conference*, Lake Buena Vista, FL, November 13–14, Society of Automotive Engineers, Warrendale PA.
[10] Van Ee, C. A., Nightingale, R. W., Camacho, D. L. A., Chancey, V. C., Knaub, K. E., Sun, E. A., and Myers, B. S., 2000, "Tensile Properties of the Human Muscular and Ligamentous Cervical Spine," Stapp Car Crash Journal, **44**, pp. 85–102.
[11] Chancey, V. C., Nightingale, R. W., Van Ee, C. A., Knaub, K. E., and Myers, B. S., 2000, "Improved Estimation of Human Neck Tensile Tolerance: Reducing the Range of Reported Tolerance Using Anthropometrically Correct Muscles and Optimized Physiologic Initial Conditions," Stapp Car Crash Journal, **47**, pp. 135–153.
[12] Blackburn, J. T., Padua, D. A., Riemann, B. L., and Guskiewicz, K. M., 2004, "The Relationships Between Active Extensibility, and Passive and Active Stiffness of the Knee Flexors," J. Electromyogr Kinesiol, **14**(6), pp. 683–691.
[13] McHugh, M. P., and Hogan, D. E., 2004, "Effect of Knee Flexion Angle on Active Joint Stiffness," Acta Physiol. Scand., **180**(3), pp. 249–254.
[14] Sinkjaer, T., Toft, E., Andreassen, S., and Hornemann, B. C., 1988, "Muscle Stiffness in Human Ankle Dorsiflexors: Intrinsic and Reflex Components," J. Neurophysiol., **60**(3), 1110–1121.
[15] Brault, J. R., Siegmund, G. P., and Wheeler, J. B., 2000, "Cervical Muscle Response During Whiplash: Evidence of a Lengthening Muscle Contraction," Clin. Biomech. (Bristol, Avon), **15**(6), pp. 426–435.
[16] Richmond, F. J. R., and Vidal, P. P., 1988, "The Motor System: Joints and Muscles of the Neck, *Control of Head Movement*, B. W. Peterson and F. J. Richmond, eds., Oxford University Press, New York.
[17] Kettler, A., Hartwig, E., Schultheiss, M., Claes, L., and Wilke, H. J., 2002, "Mechanically Simulated Muscle Forces Strongly Stabilize Intact and Injured Upper Cervical Spine Specimens," J. Biomech., **35**(3), pp. 339–346.
[18] Reber, J. G., and Goldsmith, W., 1979, "Analysis of Large Head-Neck Motions," J. Biomech., **12**(3), pp. 211–222.
[19] Luo, Z. P., and Goldsmith, W. 1991, "Reaction of a Human Head/Neck/Torso System to Shock," J. Biomech., **24**(7), pp. 499–510.
[20] Vasavada, A. N., Li, S., and Delp, S. L., 1998, "Influence of Muscle Morphometry and Moment Arms on the Moment-Generating Capacity of Human Neck Muscles," Spine, **23**(4), pp. 412–422.
[21] Brolin, K., Halldin, P., and Leijonhufvud, I., 2005, "The Effect of Muscle Activation on Neck Response," Traffic Inj Prev, **6**, pp. 67–76.
[22] Huston, R. L., Huston, J. C., and Harlow, M. W., 1978, "Comprehensive, Three-Dimensional Head-Neck Model for Impact and High-Acceleration Studies," Aviat., Space Environ. Med., **49**(1) pp. 205–210.
[23] Yang, K. H., Zhu, F., Luan, F., Zhao, L., and Begeman, P. C., 1998, "Development of a Finite Element Model of the Human Neck (983157)," *Proceedings 42nd Stapp Car Crash Conference*, Tempe, AZ, November 2–4, Society of Automotive Engineers, Warrendale PA.
[24] Stemper, B. D., Yoganandan, N., and Pintar, F. A., 2004, "Validation of a Head-Neck Computer Model for Whiplash Simulation," Med. Biol. Eng. Comput., **42**(3), pp. 333–338.
[25] Deng, Y. C., and Goldsmith, W., 1987, "Response of a Human Head/Neck/Upper-Torso Replica to Dynamic Loading—II. Analytical/Numerical Model," J. Biomech., **20**(5), pp. 487–97.
[26] Schneider, L. W., and Bowman, B. M.., 1978, "Prediction of Head/Neck Dynamic Response of Selected Military Subjects to -Gx Acceleration," Aviat., Space Environ. Med., **49**(1), 211–223.
[27] Kamibayashi, L. K., and Richmond, F. J., 1998, "Morphometry of Human Neck Muscles," Spine, **23**, pp. 1314–23.
[28] Padgaonkar, A. J., Krieger, K. W., and King, A. I., 1975, "Measurement of Angular Acceleration of a Rigid Body Using Linear Accelerometers," ASME J. Appl. Mech., **42**, pp 552–556.
[29] Blouin, J. S., Descarreaux, M., Belanger-Gravel, A., Simoneau, M., and Teasdale, N. 2003, "Attenuation of Human Neck Muscle Activity Following Repeated Imposed Trunk-Forward Linear Acceleration," Exp. Brain Res., **150**(4), pp. 458–464.
[30] Siegmund, G. P., Sanderson, D. J., Myers, B. S., and Inglis, J. T., 2003, "Rapid Neck Muscle Adaptation Alters the Head Kinematics of Aware and Unaware Subjects Undergoing Multiple Whiplash-Like Perturbations," J. Biomech., **36**(4), pp. 473–482.
[31] NASA, 1978, "Anthropometric Source Book," *Anthropometry for Designers*, NASA Reference Publication 1024, National Aeronautics and Space Administration, Scientific and Technical Information Office, Hanover, MD, Vol. 1.
[32] Siegmund, G. P., King, D. J., Lawrence, J. M., Wheeler, J. B., Brault, J. R., and Smith, T. A., 1997, "Head/Neck Kinematic Response of Human Subjects in Low-speed Rear-end Collisions (973341)," *Proceedings 41st Stapp Car Crash Conference* (P-315), Lake Buena Vista, FL, November 13–14, Society of Automotive Engineers, Warrendale, PA, pp. 357–385.
[33] Clauser, C. E., McConville, J. T., and Young, J. W., 1969, "Weight, Volume, and Center of Mass of Segments of the Human Body," Wright Patterson Air Force Base, Aerospace Medical Research Laboratory, Yellow Springs, OH, Paper No. AMRL-TR-69-70.
[34] ATA Sensors, 2005, "Application Not AN-01 (Revision A) Extending the Frequency Response of the, MHD Angular Rate Sensor," http://www.atasensors.com/Sensors2/pdf_files/Application Note AN-01 RevA.pdf downloaded May, 2005.
[35] Vasavada, A. N., Peterson, B. W., Delp, and S. L., 2002, "Three-Dimensional Spatial Tuning of Neck Muscle Activation in Humans," Exp. Brain Res., **147**, pp. 437–448.
[36] Gabriel, D. A., Matsumoto, J. Y., Davis, D. H., Currier, B. L., and An, K. N., 2004, "Multidirectional Neck Strength and Electromyographic Activity for Normal Controls," Clin. Biomech. (Bristol, Avon), **19**, pp. 653–658.
[37] Keshner, E. A., Campbell, D., Katz, R. T., and Peterson, B. W., 1989, "Neck Muscle Activation Patterns in Humans During Isometric Head Stabilization," Exp. Brain Res., **75**, pp. 335–344.
[38] Mayoux-Benhamou, M. A., Revel, M., and Vallee, C., 1997, "Selective Electromyography of Dorsal Neck Muscles in Humans," Exp. Brain Res., **113**, pp. 353–360.
[39] Williams, P. L., 1995, *Gray's Anatomy*, 38th ed., Churchill Livingstone, Edinburgh, U.K..
[40] Siegmund, G. P., Sanderson, D. J., Myers, B. S., and Inglis, J. T., 2003, "Awareness Affects the Response of Human Subjects Exposed to a Single Whiplash-Like Perturbation," Spine, **28**, pp. 671–679.
[41] Vitti, M., Fujiwara, M., Basmajian, J. M., and Iida, M., 1973, "The Integrated Roles of Longus Colli and Sternocleidomastoid Muscles: An Electromyographic Study," Anat. Rec., **177**, pp. 471–484.
[42] Kumar, S., Narayan, Y., and Amell, T., 2002, "An Electromyographic Study of Low-Velocity Rear-End Impacts," Spine, **27**, pp. 1044–1055.
[43] Vasavada, A. N., Li, S., Delp, S. L., 2001, "Three-Dimensional Isometric Strength of Neck Muscles in Humans," Spine **26**, pp. 1904–1909.
[44] Winkelstein, B. A., McLendon, R. E., Barbir, A., and Myers, B. S., 2001, "An Anatomic Investigation of the Human Cervical Facet Capsule Quantifying Muscle Insertion Area," J. Anat., **198**, pp. 455–461.
[45] Anderson, J. S., Hsu, A. W., and Vasavada, A. N., 2005, "Morphology, Architecture, and Biomechanics of Human Cervical Multifidus," Spine, **30**, pp. E86–E91.
[46] Winkelstein, B. A., Nightingale, R. W., Richardson, W. J., and Myers, B. S., 2000, "The Cervical Facet Capsule and Its Role in Whiplash Injury: A Biomechanical Investigation," Spine, **25**, pp. 1238–246.
[47] Siegmund, G. P., Myers, B. S., Davis, M. B., Bohnet, H. F., and Winkelstein, B. A., 2001, "Mechanical Evidence of Cervical Facet Capsule Injury During Whiplash: A Cadaveric Study Using Combined Shear, Compression, and Extension Loading," Spine, **26**, pp. 2095–2101.
[48] Siegmund, G. P., Heinrichs, B. E., Lawrence, J. M., and Phillipens, M. M.G.M., 2001, "Kinetic and Kinematic Responses of the RID2a, Hybrid III and Human Volunteers in Low-speed Rear-end Collisions," *Proceedings 45th Stapp. Car Crash Conference*, Stapp Association, Ann Arbor, MI, pp. 239–256.
[49] Siegmund, G. P., Sanderson, D. J., and Inglis, J. T., 2004, "Gradation of Neck Muscle Responses and Head/Neck Kinematics to Acceleration and Speed Change in Rear-End Collisions," Stapp Car Crash Journal, **48**, pp. 419–430.

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